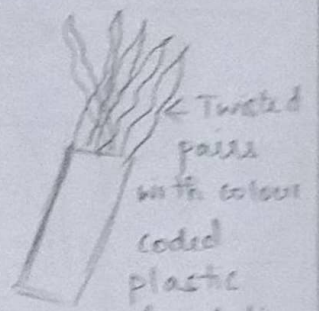
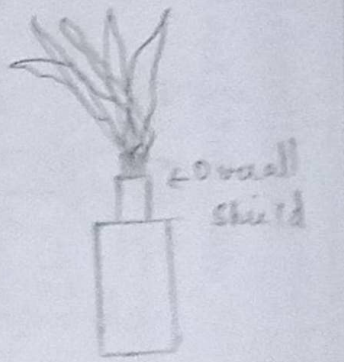
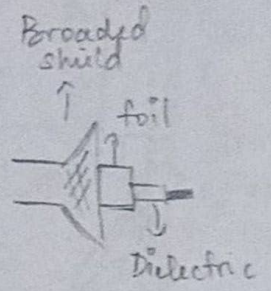
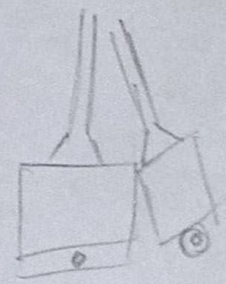
Exp no: 02 Date: 27/7/20	Study of different types of Network Cables.

Understand different types of network cables different type of cables used in networking are:

1. Unshield Twisted Pair (UTP) cable
2. Shield Twisted Pair (STP) cable
3. Coaxial cable
4. Fibre Optic cable.

Cable Type	Category	Maximum data Transmission	Advantage / Disadvantage	Application	Image
UTP	Category 3	10 bps	Adv: * Cheaper in cost * Easy to install as they have a smaller overall diameter Disadv: * More prone to (EMI) Electromagnetic interference and noise.	10 Base - T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet.	
	Category 5	upto 100Mpbs			
	Category 5e	1Gbps			
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded * Faster than UTP * Less susceptible to noise & interference. Disadv: * Expensive * Greater Installation * effect	• Gigabit Ethernet, • 10 G Ethernet (55m) Widely used in data centers	
SSTP	Category 7	10 Gbps.		Gigabit Ethernet, 10 G Ethernet	

Coaxial cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Adv: • High bandwidth • Immune to interference. • Low loss bandwidth • Versatile. Disadv: • Limited distance • Cost • Size is bulky	Speed of signal is 500m, Television network, High speed internet	
fibre optics cable.	Single mode Multi mode	100Gbps	Adv: • High speed • High bandwidth • High security • Long distance. Disadv: • Expensive • Requires skilled installer	Maximum distance of fibre optics cable is around 100 meters.	

Result:

The types of network cables and making cross wired cable & straight through cable using clamping / crimping tool is studied.